

Logistic Regression Analysis of California Homicide Cases in 2014

by

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Final Project for STA 6238 Logistic Regression  
Spring 2026

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May 2, 2026

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## **SECTION I: INTRODUCTION**

Homicide case resolution is an important public safety and criminal justice issue because solved cases provide answers for victims' families, support accountability, and help communities understand patterns of violence. One factor that may be related to whether a case is solved is the relationship between the victim and perpetrator. Cases involving family members, friends, intimate partners, or acquaintances may differ from stranger-related cases because the victim-perpetrator connection can affect the direction and difficulty of an investigation.

Other case and victim characteristics may also be associated with case resolution, including weapon type, victim age, victim sex, victim race, and agency type. This analysis uses California homicide data from 2014 to examine factors associated with whether a case was solved. The binary outcome variable is Crime Solved, coded as Yes or No, making logistic regression an appropriate modeling approach.

The analysis begins with exploratory data analysis to examine missing values, outliers, and distributions of the outcome and predictors. Relationship and weapon type are recoded into broader categories to improve interpretability and reduce sparse cells. Bivariate analyses are then used to evaluate each predictor before fitting a multiple logistic regression model. Model refinement considers statistical significance, interpretability, possible interaction terms, and diagnostics. The final selected model is evaluated using parameter estimates, odds ratios, classification performance, goodness-of-fit measures, and the ROC curve.

## **SECTION II: EXPLORATORY DATA ANALYSIS**

The dataset contained 1,872 observations. During exploratory analysis, five observations had Victim Age coded as 998, which appeared to be a placeholder rather than a valid age. These values were recoded as missing, leaving 1,867 observations for logistic regression because SAS excludes observations with missing values in model variables.

The outcome variable was Crime Solved, coded as Yes or No. The candidate predictors were Victim Age, Victim Sex, Victim Race, Agency Type, Relationship, and Weapon. Victim Age was treated as quantitative, while the remaining predictors were categorical.

Relationship and Weapon were recoded to improve interpretability and reduce sparse categories. Relationship was recoded into Relationship2, with Unknown retained as its own category because it has a different meaning from Stranger. Weapon was recoded into Weapon2 with four categories: Firearm, Knife, Unknown, and Other. Details of the recoding are shown in the appendix.

Bivariate analyses were conducted to examine the association between each predictor and Crime Solved. Chi-square tests were used for categorical predictors, while Victim Age was examined using summary statistics, outlier checks, and a bivariate logistic regression model. Victim Sex, Victim Race, Relationship2, and Weapon2 were significantly associated with Crime Solved, with chi-square p-values less than 0.0001. Agency Type was not statistically significant,  $p = 0.4028$ , but was still considered in the initial multivariable model because bivariate tests do not adjust for other predictors. The full summary of categorical bivariate results is shown in Appendix Table A1.

Relationship2 showed the strongest bivariate association with Crime Solved. The cross-tabulation is shown in Appendix Table A2. The key pattern was that 776 of 779 unsolved cases had unknown relationship status, while the other relationship categories were almost always solved. This suggests quasi-complete separation and supports treating Relationship2 cautiously during model building. In addition, Relationship2 = Unknown may be a consequence of the case being unsolved rather than an independent predictor, because relationship information may depend on whether the perpetrator was identified.

Based on the exploratory and bivariate analyses, the initial full logistic regression model included Victim Age, Victim Sex, Victim Race, Agency Type, Relationship2, and Weapon2. This allowed each predictor to be examined while adjusting for the other variables in the model.

### **SECTION III: MODEL BUILDING AND FINAL LOGISTIC REGRESSION MODEL**

A full main-effects logistic regression model was first fit using Crime Solved as the binary outcome. The initial model included Victim Age, Victim Sex, Victim Race, Agency Type, Relationship2, and Weapon2. In this model, Relationship2 was statistically significant, but the parameter estimates showed evidence of quasi-complete separation. Because relationship status may depend on whether the perpetrator was identified, Relationship2 may not function as an independent predictor of case resolution. Therefore, Relationship2 was excluded from the final model.

After excluding Relationship2, the model was refit with Victim Age, Victim Sex, Victim Race, Agency Type, and Weapon2. Agency Type was not statistically significant,  $p = 0.8092$ , and was removed. Interaction terms were also examined. Although some joint interaction tests were statistically significant, follow-up frequency tables showed sparse cells and unstable estimates, especially for some race-weapon combinations. Therefore, interaction terms were interpreted cautiously and were not retained.

The final logistic regression model included Victim Age, Victim Sex, Victim Race, and Weapon2. Victim Age was retained because it was significant in the bivariate model and is a meaningful demographic predictor, even though it was not statistically significant at the 0.05 level in the adjusted final model. Male, White, and Firearm were used as the reference categories for Victim Sex, Victim Race, and Weapon2, respectively.

The final model can be written in the general form:

$$\log \left( \frac{\hat{\pi}}{1 - \hat{\pi}} \right) = \beta_0 + \beta_1(\text{Victim Age}) + \beta_2(\text{Victim Sex}) + \beta_3(\text{Victim Race}) + \beta_4(\text{Weapon2})$$

where Victim Sex, Victim Race, and Weapon2 are represented by indicator variables with Male, White, and Firearm as the reference categories.

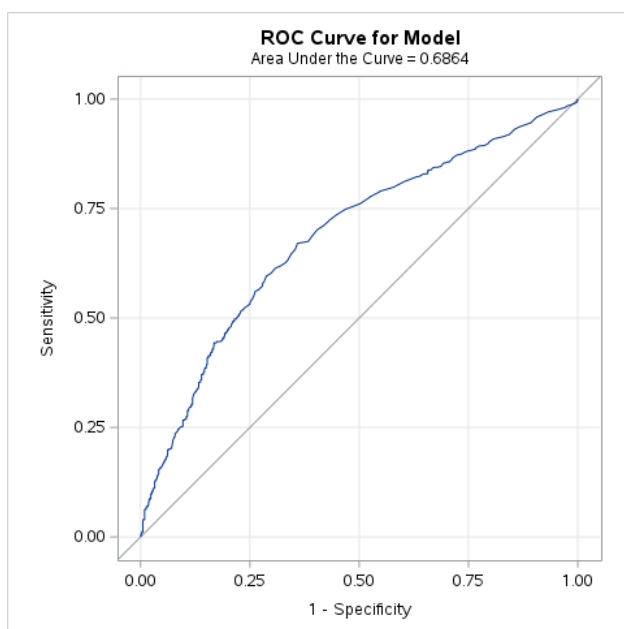
| Predictor                                          | Coeff Estimate | OR     | 95% OR CI |       | P-value |
|----------------------------------------------------|----------------|--------|-----------|-------|---------|
| Victim Age                                         | 0.0059         | 1.0060 | 0.999     | 1.012 | 0.0709  |
| Victim Sex Female vs Male                          | 1.0019         | 2.7230 | 2.012     | 3.686 | <0.0001 |
| Victim Race Asian/Pacific Islander vs White        | 0.2176         | 1.2430 | 0.736     | 2.100 | 0.4163  |
| Victim Race Black vs White                         | -0.4287        | 0.6510 | 0.525     | 0.808 | <0.001  |
| Victim Race Native American/Alaska Native vs White | -0.0957        | 0.9090 | 0.285     | 2.901 | 0.8716  |
| Victim Race Unknown vs White                       | -0.4864        | 0.6150 | 0.323     | 1.171 | 0.1390  |
| Weapon2 Knife vs Firearm                           | 1.1435         | 3.1380 | 2.259     | 4.358 | <0.001  |
| Weapon2 Other vs Firearm                           | 0.7828         | 2.1880 | 1.543     | 3.102 | <0.001  |
| Weapon2 Unknown vs Firearm                         | 0.0173         | 1.0170 | 0.609     | 1.700 | 0.9474  |

*Table 1. Final Logistic Regression Model Results*

The final model showed that Victim Sex, Victim Race, and Weapon2 were statistically significant predictors of case resolution. Female victims had higher odds of case resolution than male victims. Compared with White victims, Black victims had significantly lower odds of case resolution. Compared with firearm cases, cases involving knives or other weapons had higher

odds of being solved. Victim Age had a positive coefficient but was not statistically significant in the adjusted model,  $p = 0.0709$ .

At the 0.50 probability cutoff, the model correctly classified 62.7% of cases, with sensitivity of 82.2% and specificity of 35.2% (Appendix Table A6). This indicates that the model was better at identifying solved cases than unsolved cases. Model diagnostic plots were also examined to assess residuals, leverage, and influence (Appendix Figure A1). Several observations showed relatively large residuals or influence, but no observations were removed because there was no evidence that they represented data entry errors.



*Figure 1. ROC Curve for Final Logistic Regression Model*

The ROC curve shows the model's ability to distinguish between solved and unsolved cases across all probability cutoffs. The AUC was 0.6864, indicating modest discriminatory ability. The model performed better than chance but did not strongly separate solved from unsolved cases, which is consistent with the classification results.

Overall, the final model balanced statistical significance, interpretability, and model stability. Relationship2 was excluded due to quasi-complete separation and possible dependence on the outcome, Agency Type was excluded due to lack of statistical significance, and interaction terms were excluded because of sparse-cell instability.

#### SECTION IV: AI-GENERATED ANALYSIS AND COMPARISON

This section presents a fully AI-generated version of the analysis using Codex. The exact prompt and full AI response are included in Appendix Part II.

The AI-generated analysis and my analysis followed a similar overall workflow. Both imported the dataset, examined variable structure and frequency distributions, identified Victim Age = 998 as an unusual placeholder value, recoded it as missing, and fit logistic regression models for the binary outcome Crime Solved. Both analyses also recognized that some categorical variables should be collapsed because of sparse or overlapping categories.

The main differences were in variable selection and model refinement. The AI-generated analysis created a numeric outcome variable, Solved, while my analysis modeled the original Crime Solved variable directly using event='Yes'. The AI version did not include Relationship; my assumption is that it excluded this variable because many relationship categories were sparse and because unknown relationship status was closely tied to whether the case was solved. In contrast, my analysis examined Relationship because of its importance in homicide investigations. I found that Relationship2 was strongly associated with case resolution, but excluded it from the final model because it showed quasi-complete separation and possible dependence on the outcome.

The AI-generated analysis also collapsed Victim Race and Agency Type into broader categories, while my analysis kept the original categories and removed Agency Type only after evaluating it in the multivariable model. My analysis also included interaction assessment, sparse-cell checks, ROC interpretation, classification performance, and diagnostic plots. Overall, the AI version produced a reasonable simplified model, while my final analysis was more detailed and more closely aligned with the course model-building workflow.

The AI-generated final model produced conclusions similar to my final model for shared predictors. Both models found higher odds of case resolution for female victims and for non-firearm weapons such as knives, and lower odds for Black victims compared with White victims. For example, the AI estimated an odds ratio of 2.851 for Female vs Male, compared with 2.723 in my model, and an odds ratio of 0.638 for Black vs White, compared with 0.651 in my model.

The standard errors for comparable predictors were also similar, although some estimates were not directly comparable because the AI collapsed Victim Race and Weapon differently. The AI model had modest discrimination, with  $c = 0.671$ , slightly lower than the AUC of 0.6864 from my final model. Its Hosmer-Lemeshow p-value was 0.2071, suggesting no strong evidence of poor fit.

Overall, the AI's odds ratio interpretations were generally statistically accurate, but its reasoning was more generic. In particular, it did not investigate Relationship as a substantively important predictor, did not identify the quasi-complete separation issue, and did not assess interaction terms. For supporting details, see Tables B1-B3 in Appendix Part II.

## APPENDIX

### Appendix Part I: My Version Analysis Supplements

Table A1. Summary of Categorical Predictors by Crime Solved

| Predictor     | Categories                                                                   | Chi-Square<br>p-value | EDA Conclusion                                   |
|---------------|------------------------------------------------------------------------------|-----------------------|--------------------------------------------------|
| Victim Sex    | Male, Female                                                                 | <0.001                | Associated with Crime Solved                     |
| Victim Race   | Asian/Pacific Islander, Black, Native American/Alaska Native, Unknown, White | <0.001                | Associated with Crime Solved                     |
| Agency Type   | Municipal Police, Sheriff, Special Police                                    | 0.4028                | Not significant in bivariate analysis            |
| Relationship2 | Family, Friend, Intimate Partner, Stranger, Unknown                          | <0.001                | Strong association but possible separation issue |
| Weapon2       | Firearm, Knife, Other, Unknown                                               | <0.001                | Associated with Crime Solved                     |

Table A2. Relationship2 by Crime Solved

| Relationship2    | Not Solved | Solved       | Total        | Percent Solved |
|------------------|------------|--------------|--------------|----------------|
| Family           | 1          | 154          | 155          | 99.35%         |
| Friend           | 0          | 197          | 197          | 100.00%        |
| Intimate Partner | 0          | 130          | 130          | 100.00%        |
| Stranger         | 2          | 435          | 437          | 99.54%         |
| Unknown          | 776        | 177          | 953          | 18.57%         |
| <b>Total</b>     | <b>779</b> | <b>1,093</b> | <b>1,872</b> | <b>58.39%</b>  |



Table A3. Relationship Recoding Scheme

| <b>Original Relationship Levels</b>                                                                      | <b>Recoded Category</b> |
|----------------------------------------------------------------------------------------------------------|-------------------------|
| Boyfriend, Boyfriend/Girlfriend, Girlfriend, Husband, Wife, Common-Law Husband, Common-Law Wife, Ex-Wife | Intimate Partner        |
| Brother, Daughter, Family, Father, In-Law, Mother, Sister, Son, Stepfather, Stepson                      | Family                  |
| Friend, Acquaintance, Neighbor, Employer                                                                 | Friend                  |
| Stranger                                                                                                 | Stranger                |
| Unknown                                                                                                  | Unknown                 |

Table A4. Weapon Recoding Scheme

| <b>Original Weapon Levels</b>                                   | <b>Recoded Category</b> |
|-----------------------------------------------------------------|-------------------------|
| Firearm, Gun, Handgun, Rifle, Shotgun                           | Firearm                 |
| Knife                                                           | Knife                   |
| Unknown, blank/missing                                          | Unknown                 |
| Blunt Object, Drowning, Drugs, Fire, Strangulation, Suffocation | Other                   |

Table A5. Interaction Assessment Summary

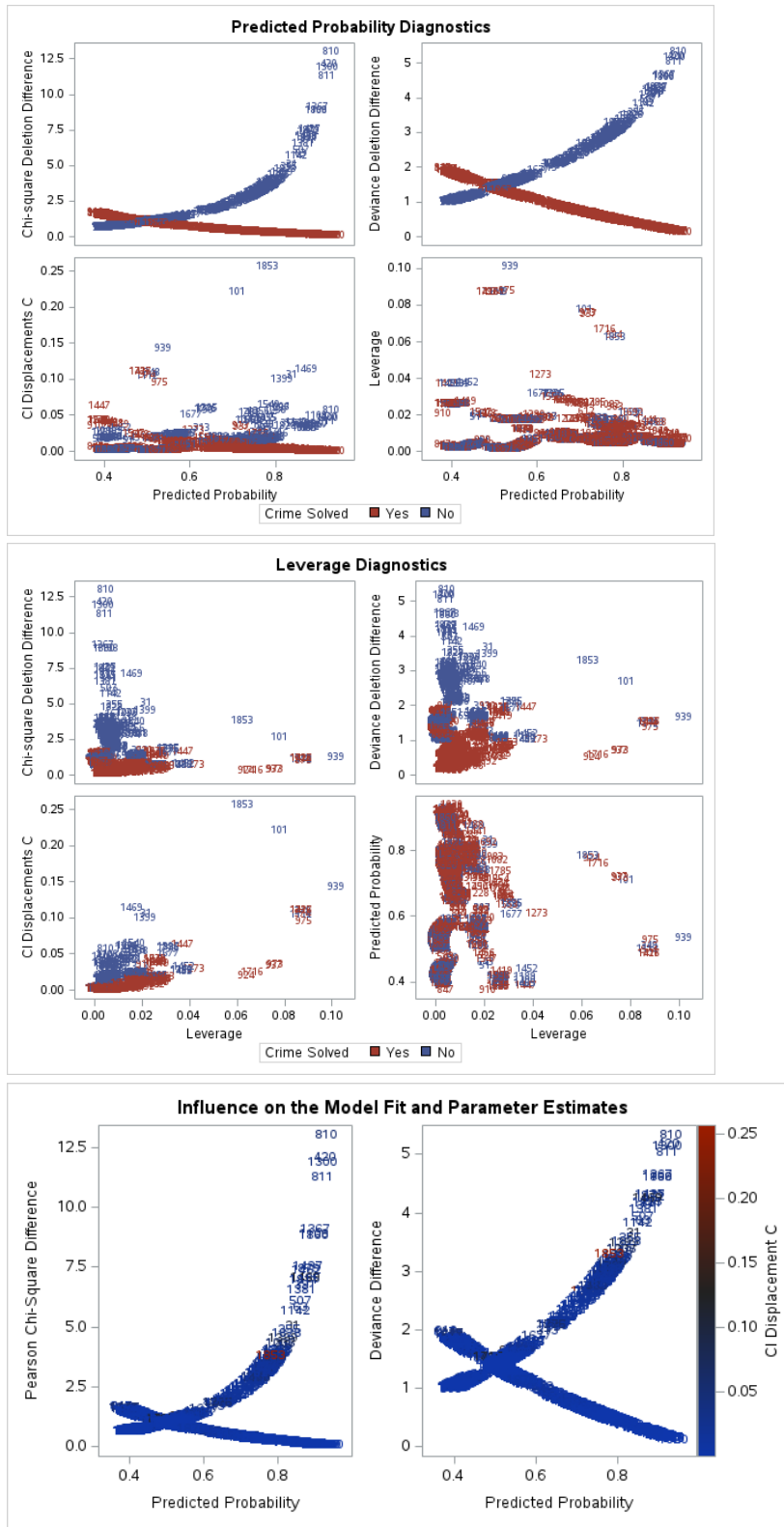
| <b>Interaction</b>              | <b>p-value</b>   | <b>Decision</b>                             |
|---------------------------------|------------------|---------------------------------------------|
| Victim Age $\times$ Victim Sex  | 0.0602           | Not retained                                |
| Victim Age $\times$ Victim Race | 0.1364           | Not retained                                |
| Victim Sex $\times$ Victim Race | 0.0347           | Not retained due to sparse-cell instability |
| Victim Sex $\times$ Weapon2     | 0.4517           | Not retained                                |
| Victim Race $\times$ Weapon2    | 0.0045 (reduced) | Not retained due to sparse cells            |

Table A6. Confusion Matrix for Final Logistic Regression Model at Probability Cutoff 0.5

| Actual Crime Solved | Predicted Yes | Predicted No | Total |
|---------------------|---------------|--------------|-------|
| Yes                 | 898           | 194          | 1,092 |
| No                  | 502           | 273          | 775   |
| Total               | 1,400         | 467          | 1,867 |

Sensitivity = 82.2%; Specificity = 35.2%

Figure A1. Predicted Probability Diagnostics, Leverage Diagnostics, and Influence on the Model Fit and Parameter Estimates.



## SAS Code (my version)

```
/*STA6238 Final Project - Student Version
Name: Xiaoling Sundberg
Date: April 26, 2026*/

/*Read in dataset - 'homicide_california_2014.csv'
this is a subset of the homicide.csv dataset that contains over
630,000 observations.
The original dataset includes all homicide cases across 50 states in
the US from 1980 to 2014.
For this analysis, I chose the state of California in year 2014 be-
cause it is a big state and has great ethnic diversity.
The binary target variable is 'Crime Solved' with outcome 'Yes' and
'No'
The analysis dataset contains 1,872 observations.*/

*Read in Dataset;
proc import datafile="/home/u64397595/STA 6238/homicide_californi-
a_2014.csv"
    out=homicide_california_2014
    dbms=csv
    replace;
    guessingrows=max;
run;
proc print data=homicide_california_2014 (obs=10);
run;

/*****
SECTION I: EXPLORATORY DATA ANALYSIS.
*****/

*1.Select Predictors;
data crime_analysis;
    set homicide_california_2014;

    Obs = _N_;

    keep Obs
        'Crime Solved'n
        'Victim Sex'n
        'Victim Race'n
        'Victim Age'n
        'Agency Type'n
        Relationship
        Weapon;
run;
proc print data=crime_analysis (obs=10);
run;
```

```

*2.Recode the categorical predictors 'Relationship' and 'Weapon';
*Relationship original levels - using SQL;
proc sql;
    select distinct Relationship
    from crime_analysis
    order by Relationship;
quit;

*Recode Relationship;
data crime_analysis;
    set crime_analysis;

    length Relationship2 $25;

    if Relationship in ("Boyfriend", "Boyfriend/Girlfriend", "Girl-
friend",
                        "Husband", "Wife", "Common-Law Husband", "Com-
mon-Law Wife",
                        "Ex-Wife") then Relationship2 = "Intimate
Partner";

    else if Relationship in ("Brother", "Daughter", "Family",
"Father",
                        "In-Law", "Mother", "Sister", "Son",
                        "Stepfather", "Stepson") then Relation-
ship2 = "Family";

    else if Relationship in ("Friend", "Acquaintance", "Neighbor",
"Employer")
        then Relationship2 = "Friend";

    else if Relationship = "Stranger" then Relationship2 = "Stranger";
    else if Relationship = "Unknown" then Relationship2 = "Unknown";

    else Relationship2 = "Other";
run;

*Check recoded relationship levels;
proc freq data=crime_analysis;
    tables Relationship2 / missing;
run;

*Compare original and recoded relationship variables;
proc freq data=crime_analysis;
    tables Relationship*Relationship2 / missing;
run;

*Weapon original levels - using SQL;
proc sql;
    select distinct Weapon
    from crime_analysis
    order by Weapon;
quit;

```

```

*Recode Weapon;
data crime_analysis;
    set crime_analysis;

    length Weapon2 $20;

    if Weapon in ("Firearm", "Gun", "Handgun", "Rifle", "Shotgun")
then Weapon2 = "Firearm";
    else if Weapon = "Knife" then Weapon2 = "Knife";
    else if Weapon in ("Unknown", "") then Weapon2 = "Unknown";
    else Weapon2 = "Other";
run;

*Check recoded weapon levels;
proc sql;
    select distinct Weapon2
    from crime_analysis
    order by Weapon2;
quit;

*Compare original and recoded weapon variables;
proc freq data=crime_analysis;
    tables Weapon*Weapon2 / missing;
run;

*Check selected and recoded variables;
proc print data=crime_analysis(obs=20);
    var 'Crime Solved'n
        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        'Agency Type'n
        Relationship2
        Weapon2;
run;

*3.Victim Age is a quantitative predictor variable. Check outliers;
proc univariate data=crime_analysis nextrobs=10;
    var 'Victim Age'n;
run;

*Found age 998, which appears to be a placeholder. Recode it as miss-
ing;
data crime_analysis;
    set crime_analysis;

    if 'Victim Age'n = 998 then 'Victim Age'n = .;
run;

*Re-run after recoding age 998 as missing;
proc univariate data=crime_analysis nextrobs=10;
    var 'Victim Age'n;
run;
*The new highest age value is now 96;

```

```

/* There is also age of zero as the lowest value.
Since age zero can represent an infant victim, keep age zero as valid.
There are 5 missing values after recoding 998 as missing. */
proc freq data=crime_analysis;
    tables 'Victim Age'n / missing;
run;

*Exploratory scatter plot of Victim Age vs. Crime Solved;
proc sgplot data=crime_analysis;
    yaxis label="Crime Solved";
    xaxis label="Victim Age" values=(0 to 100 by 10);
    scatter x='Victim Age'n y='Crime Solved'n;
run;

/*4. Univariate and bivariate analysis:
Examine the relationship between each predictor and Crime Solved */

/* Categorical predictors */
proc freq data=crime_analysis;
    tables 'Crime Solved'n*'Victim Sex'n / chisq;
    tables 'Crime Solved'n*'Victim Race'n / chisq;
    tables 'Crime Solved'n*'Agency Type'n / chisq;
    tables 'Crime Solved'n*'Relationship2' / chisq;
    tables 'Crime Solved'n*'Weapon2' / chisq;
run;

/* Quantitative predictor: Victim Age */
proc logistic data=crime_analysis;
    model 'Crime Solved'n(event='Yes') = 'Victim Age'n;
run;
*Note:      5 observations were deleted due to missing values for the
response or explanatory variables.

/*****
SECTION II: BUILD A FULL LOGISTIC REGRESSION MODEL
*****/

*Full candidate model with all selected predictors;
proc logistic data=crime_analysis;
    class 'Victim Sex'n
        'Victim Race'n
        'Agency Type'n
        Relationship2
        Weapon2 / param=ref;

    model 'Crime Solved'n(event='Yes') =
        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        'Agency Type'n
        Relationship2
        Weapon2
        / link=logit clodds=wald;
run;

```

```

/*In the full model, Relationship2 was statistically significant;
however, the model showed evidence of quasi-complete separation,
with extremely large standard errors for some relationship categories.
Because relationship status may depend on whether the perpetrator was
identified, Relationship2 was excluded from the final model.*/

```

```

proc logistic data=crime_analysis plots(only)=roc;
  class 'Victim Sex'n
        'Victim Race'n
        'Agency Type'n
        Weapon2 / param=ref;

  model 'Crime Solved'n(event='Yes') =
        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        'Agency Type'n
        Weapon2
        / link=logit clodds=wald ctable pprob=0.5;

```

```

run;
/* After excluding Relationship2, the model no longer showed the same
separation issue. Victim Sex, Victim Race, and Weapon2 were statisti-
cally
significant predictors. Agency Type was not significant and was re-
moved
from the next model. Victim Age was not significant at the 0.05 level
but
was retained temporarily because it was significant in the bivariate
model
and remained close to significant in the multivariable model. */

```

```

*Create interaction term(s);
proc logistic data=crime_analysis;
  class 'Victim Sex'n
        'Victim Race'n
        Weapon2 / param=ref;

  model 'Crime Solved'n(event='Yes') =
        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        Weapon2
        'Victim Age'n*'Victim Sex'n
        'Victim Age'n*'Victim Race'n
        'Victim Race'n* 'Victim Sex'n
        'Victim Sex'n*Weapon2
        'Victim Race'n*Weapon2
        / link=logit clodds=wald ctable pprob=0.5;

```

```

run;
*Victim Race*Victim Sex and Victim Race*Weapon2 are significant.

```

```

*Interaction model retaining significant two-way interactions;
proc logistic data=crime_analysis;
  class 'Victim Sex'n

```



```

        'Victim Race'n
        Weapon2 / param=ref;

    model 'Crime Solved'n(event='Yes') =
        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        Weapon2
        'Victim Race'n*'Victim Sex'n
        'Victim Race'n*Weapon2
        / link=logit clodds=wald ctable pprob=0.5;
run;

/* The interaction terms were evaluated using joint Wald tests because
categorical interactions are represented by multiple parameter esti-
mates.
The joint test for Victim Race by Weapon2 was significant, although
many
individual interaction parameters were not significant. This can occur
when the overall pattern differs across categories, but no single
category-level comparison is estimated precisely. Some very large
standard
errors suggest sparse cells for certain race-weapon combinations, so
this
interaction should be interpreted cautiously. */

*Check the 3-way table;
proc freq data=crime_analysis;
    tables 'Victim Race'n*Weapon2*'Crime Solved'n / missing;
run;
/* Note on Victim Race by Weapon2 interaction:
The joint test for the Victim Race by Weapon2 interaction was statis-
tically
significant, suggesting that the association between weapon type and
case
resolution may differ by victim race. However, cross-tabulations
showed
sparse cells for some race-weapon combinations, especially for Native
American/Alaska Native victims and Unknown victim race. These sparse
cells
produced very large standard errors for some interaction parameters.
Therefore, this interaction was interpreted cautiously and was not re-
tained
in the final model.
*/

***** Final model *****;
*Final model with goodness-of-fit and classification table;
proc logistic data=crime_analysis plots(only)=roc;
    class 'Victim Sex'n(ref='Male')
        'Victim Race'n(ref='White')
        Weapon2(ref='Firearm') / param=ref;

    model 'Crime Solved'n(event='Yes') =

```

```

        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        Weapon2
        / link=logit clodds=wald ctable lackfit pprob=0.5;
run;
/* Victim Age was retained in the final model because it was statisti-
cally
significant in the bivariate logistic regression model and is a mean-
ingful
demographic predictor. Although it was not significant at the 0.05
level in
the adjusted final model (p = 0.0709), it remained close to signifi-
cance
and was kept to adjust for potential age-related differences in case
resolution. */

*Final model diagnostics;
proc logistic data=crime_analysis plots(only label)=(phat leverage
dpc);
    class 'Victim Sex'n(ref='Male')
        'Victim Race'n(ref='White')
        Weapon2(ref='Firearm') / param=ref;

    model 'Crime Solved'n(event='Yes') =
        'Victim Age'n
        'Victim Sex'n
        'Victim Race'n
        Weapon2;
run;

```

## Appendix Part II: AI-Generated Analysis Supplements

AI prompt:

*“Inspect the dataset homicide\_california\_2014.csv and build a logistic regression analysis plan for predicting Crime Solved. Decide which variables to include, which categorical variables need recoding, how to handle missing or unusual values, which bivariate analyses to run, and what final logistic regression model to use. Generate SAS code and explain the modeling decisions. Inspect the dataset from scratch and decide whether any variables should be recoded. Do not assume any predefined recoding scheme.”*

AI Full Response:

Based on the uploaded homicide\_california\_2014.csv, the dataset has 1,872 observations and 24 variables. The outcome is Crime Solved with:

- **Yes:** 1,093 cases
- **No:** 779 cases

The file includes homicide case information such as victim demographics, agency type, relationship, weapon, perpetrator information, and record identifiers.

### 1. Modeling goal

Use logistic regression to predict:

$P(\text{Crime Solved} = \text{Yes})$

$P(\text{Crime Solved} = \text{Yes})$

$P(\text{Crime Solved} = \text{Yes})$

So in SAS, code the outcome as: `Solved = (Crime_Solved = "Yes");`

The model predicts the log odds that a homicide case was solved.

### 2. Variables to exclude

Do **not** include these in the final model:

| Variable                                                                  | Reason                                                                                                    |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Record ID                                                                 | Identifier only                                                                                           |
| Agency Code, Agency Name                                                  | Too many levels, not useful for clean logistic model                                                      |
| City                                                                      | 51 levels; too many for this assignment                                                                   |
| State                                                                     | Constant; all California                                                                                  |
| Year                                                                      | Constant; all 2014                                                                                        |
| Record Source                                                             | Constant; all FBI                                                                                         |
| Incident                                                                  | Administrative count                                                                                      |
| Perpetrator Sex, Perpetrator Age, Perpetrator Race, Perpetrator Ethnicity | Strong data leakage: these are often known only when the crime is solved                                  |
| Relationship                                                              | Also creates quasi-complete separation because many known relationships occur almost only in solved cases |
| Victim Ethnicity                                                          | Closely overlaps with victim race; exclude to avoid redundancy                                            |

Important point: Perpetrator Sex, Perpetrator Race, Perpetrator Ethnicity, and Relationship are not good predictors for a fair final model because they almost directly reveal whether the case was solved.

### 3. Variables to include in the candidate model

Use these as the main candidate predictors:

| Variable          | Treatment                                                             |
|-------------------|-----------------------------------------------------------------------|
| Victim Age        | Continuous, but set 998 to missing                                    |
| Victim Sex        | Categorical                                                           |
| Victim Race       | Recode sparse levels                                                  |
| Agency Type       | Recode because Special Police is small                                |
| Weapon            | Recode into fewer categories                                          |
| Victim Count      | Numeric; optional                                                     |
| Perpetrator Count | I would not use in final because it may reflect investigation outcome |

### 4. Missing and unusual values

Victim Age

Victim Age = 998 appears to be a placeholder for unknown age. Recode it to missing.

Keep Victim Age = 0 because that likely represents infants, not missing data.

```
if Victim_Age = 998 then Victim_Age_Clean = .;  
else Victim_Age_Clean = Victim_Age;
```

Perpetrator Age

Perpetrator Age = 0 appears when the perpetrator is unknown. Since perpetrator variables leak solved status, exclude it from the final model.

### 5. Categorical recoding decisions

Agency Type

Original levels:

- Municipal Police
- Sheriff
- Special Police

Since Special Police has only 12 cases, collapse it:

```
if Agency_Type = "Municipal Police" then Agency_Type2 = "Municipal Police";  
else Agency_Type2 = "Non-Municipal";
```

Victim Race

Original levels include sparse categories. Recode as:

```
if Victim_Race in ("White", "Black", "Asian/Pacific Islander") then Victim_Race2 = Victim_Race;
else Victim_Race2 = "Other/Unknown";
```

Weapon

Collapse weapons into fewer meaningful groups:

```
if Weapon in ("Handgun", "Firearm", "Rifle", "Shotgun", "Gun") then Weapon2 = "Firearm";
else if Weapon = "Knife" then Weapon2 = "Knife";
else if Weapon = "Blunt Object" then Weapon2 = "Blunt Object";
else if Weapon = "Unknown" then Weapon2 = "Unknown";
else Weapon2 = "Other";
```

6. Bivariate analyses to run

Run these before the logistic model.

For categorical predictors

Use PROC FREQ with chi-square tests:

```
tables Solved*(Victim_Sex Victim_Race2 Agency_Type2 Weapon2) / chisq;
```

These tell whether each categorical predictor is associated with whether the case was solved.

For continuous predictors

Use PROC MEANS or PROC TTEST for victim age:

```
proc means data=homicide_clean mean std min q1 median q3 max;
  class Solved;
  var Victim_Age_Clean;
run;
```

```
proc ttest data=homicide_clean;
  class Solved;
  var Victim_Age_Clean;
run;
```

7. Suggested final model

I would fit a full candidate model first:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 (\text{Victim Age}) + \beta_2 (\text{Victim Sex}) + \beta_3 (\text{Victim Race}) + \beta_4 (\text{Agency Type}) + \beta_5 (\text{Weapon})$$

Then remove non-significant variables based on Type 3 tests and likelihood ratio comparisons.

From the data inspection, the strongest practical final model is:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 (\text{Victim Sex}) + \beta_2 (\text{Victim Race2}) + \beta_3 (\text{Weapon2})$$

where:  $\pi_i = P(\text{Crime Solved}_i = \text{Yes})$

I would remove Agency Type because it does not appear strongly associated with solved status after inspection. I would also remove Victim Age from the final parsimonious model if the professor wants only significant predictors, but I would still show it in the full model first.

#### SAS Code

```
/* Import homicide data */
proc import datafile="/home/u64397595/STA 6238/homicide_california_2014.csv"
    out=homicide
    dbms=csv
    replace;
    guessingrows=max;
run;

/* Check variable structure */
proc contents data=homicide;
run;

/* Frequency checks for categorical variables */
proc freq data=homicide;
    tables
        'Crime Solved'n
        'Agency Type'n
        'Victim Sex'n
        'Victim Race'n
        'Victim Ethnicity'n
        'Perpetrator Sex'n
        'Perpetrator Race'n
        Relationship
        Weapon
        / missing;
run;

/* Check numeric variables */
proc means data=homicide n mean std min q1 median q3 max;
    var 'Victim Age'n 'Perpetrator Age'n 'Victim Count'n 'Perpetrator Count'n;
run;

/* Clean and recode variables */
data homicide_clean;
    set homicide;
```

```

/* Outcome: 1 = solved, 0 = not solved */
if 'Crime Solved'n = "Yes" then Solved = 1;
else if 'Crime Solved'n = "No" then Solved = 0;

/* Clean victim age */
if 'Victim Age'n = 998 then Victim_Age_Clean = .;
else Victim_Age_Clean = 'Victim Age'n;

/* Recode Agency Type */
length Agency_Type2 $20;
if 'Agency Type'n = "Municipal Police" then Agency_Type2 =
"Municipal Police";
else Agency_Type2 = "Non-Municipal";
/* Recode Victim Race */
length Victim_Race2 $30;
if 'Victim Race'n in ("White", "Black", "Asian/Pacific Is-
lander") then
    Victim_Race2 = 'Victim Race'n;
else Victim_Race2 = "Other/Unknown";

/* Recode Weapon */
length Weapon2 $20;
if Weapon in ("Handgun", "Firearm", "Rifle", "Shotgun",
"Gun") then Weapon2 = "Firearm";
else if Weapon = "Knife" then Weapon2 = "Knife";
else if Weapon = "Blunt Object" then Weapon2 = "Blunt Ob-
ject";
else if Weapon = "Unknown" then Weapon2 = "Unknown";
else Weapon2 = "Other";
run;

/* Check recoded variables */
proc freq data=homicide_clean;
    tables Solved Agency_Type2 Victim_Race2 Weapon2 / missing;
run;

/* Bivariate analysis: categorical variables */
proc freq data=homicide_clean;

```

```

    tables Solved*Agency_Type2 / chisq;
    tables Solved*'Victim Sex'n / chisq;
    tables Solved*Victim_Race2 / chisq;
    tables Solved*Weapon2 / chisq;
run;

/* Bivariate analysis: victim age */
proc means data=homicide_clean n mean std min q1 median q3 max;
    class Solved;
    var Victim_Age_Clean;
run;

proc ttest data=homicide_clean;
    class Solved;
    var Victim_Age_Clean;
run;

/* Full candidate logistic regression model */
proc logistic data=homicide_clean descending;
    class
        Agency_Type2 (param=ref ref="Municipal Police")
        'Victim Sex'n (param=ref ref="Male")
        Victim_Race2 (param=ref ref="White")
        Weapon2 (param=ref ref="Firearm");
    model Solved =
        Victim_Age_Clean
        Agency_Type2
        'Victim Sex'n
        Victim_Race2
        Weapon2
        / clodds=wald;
run;

/* Reduced final logistic regression model */
proc logistic data=homicide_clean descending;
    class
        'Victim Sex'n (param=ref ref="Male")
        Victim_Race2 (param=ref ref="White")

```



```

    Weapon2 (param=ref ref="Firearm");
model Solved =
    'Victim Sex'n
    Victim_Race2
    Weapon2
    / clodds=wald lackfit;
run;

```

Final explanation to write

A good summary would be:

The final logistic regression model predicts whether a homicide case was solved using victim sex, recoded victim race, and recoded weapon type. Perpetrator-related variables were excluded because they are closely tied to whether the case was solved and would create data leakage. Relationship was also excluded because unknown relationship status is highly associated with unsolved cases and causes quasi-complete separation. Victim age was cleaned by treating age 998 as missing. Sparse categorical variables were collapsed to avoid unstable estimates. The final model is based on variables available from the victim and case characteristics rather than information that depends on identifying the perpetrator.

Table B1. AI Final Model Results

| Predictor                                    | Coefficient Estimate | Odds Ratio | 95% CI for Odds Ratio |       | P-value |
|----------------------------------------------|----------------------|------------|-----------------------|-------|---------|
| Victim Sex Female vs Male                    | 1.0478               | 2.8510     | 2.108                 | 3.857 | <0.0001 |
| Victim_Race2 Asian/Pacific Islander vs White | 0.2038               | 1.2260     | 0.725                 | 2.073 | 0.4467  |
| Victim_Race2 Black vs White                  | -0.4491              | 0.6380     | 0.515                 | 0.791 | <0.001  |
| Victim_Race2 Other/Unknown vs White          | -0.5491              | 0.5770     | 0.335                 | 0.997 | 0.0486  |
| Weapon2 Blunt Object vs Firearm              | 0.9450               | 2.5730     | 1.746                 | 3.792 | <0.001  |
| Weapon2 Knife vs Firearm                     | 1.1813               | 3.2580     | 2.349                 | 4.520 | <0.001  |
| Weapon2 Other vs Firearm                     | 0.2367               | 1.2670     | 0.645                 | 2.491 | 0.4924  |
| Weapon2 Unknown vs Firearm                   | 0.0565               | 1.0580     | 0.634                 | 1.767 | 0.8291  |

Table B2. Association of Predicted Probabilities and Observed Responses for AI Model

|                    |         |           |       |
|--------------------|---------|-----------|-------|
| Percent Concordant | 57.1    | Somer's D | 0.341 |
| Percent Discordant | 23.0    | Gamma     | 0.426 |
| Percent Tied       | 19.9    | Tau-a     | 0.166 |
| Pairs              | 851,447 | c         | 0.671 |

Table B3. Hosmer-Lemeshow Goodness-of-Fit Test for AI-Generated Model

|            |    |            |
|------------|----|------------|
| Chi-Square | DF | Pr > ChiSq |
| 8.4472     | 6  | 0.2071     |